WINDOWS: VISUAL STUDIO AND QT CREATOR

Two mainstream options, one compiler in common

TWO IDES, ONE THEME

Your choice on Windows

- **Visual Studio** - free Community edition, ships its own compiler (MSVC).
- **Qt Creator** - cross-platform IDE. It can use different compilers, including MSVC, MinGW, and LLVM/Clang. You choose a compiler when you configure a **Kit**.

Both support CMake and will read the same `CMakeLists.txt` from this course. Use what works best for you.

VISUAL STUDIO

Installing Visual Studio

1. Download the Visual Studio Installer, choose **Community** (free).
2. Under **Workloads**, check **Desktop development with C++**.
3. That workload includes MSVC, the Windows SDK, and CMake support - you don't install those separately.

OPENING A LECTURE FOLDER

Open any lecture folder directly

Visual Studio can open a folder containing a `CMakeLists.txt` without a `.sln` file at all: **File -> Open -> XXX...** and pick a lecture's folder.

Visual Studio detects `CMakeLists.txt`, configures the project, and shows `rooster` as the run/debug target.

MSVC

MSVC - the compiler that comes with it

Visual Studio's C++ compiler is called **MSVC**. Its C++23 support is controlled by a compiler flag, not a version number you pick yourself:

```
/std:c++latest
```

CMake's `set(CMAKE_CXX_STANDARD 23)` (already in this course's `CMakeLists.txt`) asks CMake to add the right flag for you automatically.

QT CREATOR

Installing Qt Creator

Use the **Qt Online Installer** to install Qt and Qt Creator.

KITS

Kits - a bundle Qt Creator needs to build anything

A **Kit** = a compiler + a debugger + (optionally) a Qt version, bundled together under one name. Qt Creator can't build a project until at least one Kit is configured.

On Windows, the installer usually detects one automatically if Visual Studio is already installed (an **MSVC kit**). You can also add:

- a **MinGW** kit (a GCC build for Windows)
- an **LLVM/Clang** kit, if you've installed one separately

PICKING A KIT

Opening this lecture in Qt Creator

File -> Open File or Project, choose this folder's `CMakeLists.txt`. Qt Creator shows a **Configure Project** screen listing every Kit it knows about - check the one you want to build with, then **Configure Project**.

DIFFERENT KITS, DIFFERENT SUPPORT

Not all kits support C++23 equally

MSVC, MinGW, and LLVM/Clang kits are three different compilers, each with its own pace of adding new C++ features. A kit with a compiler that's a version or two behind may not support everything this course uses yet.

If the compiler on your kit doesn't support the C++23 features used in this course, don't panic. I will show you how to use **Docker** to get latest compilers running on your machine.